

Module 4

3 mark

1. Describe the use of value engineering in the design process.
2. How does intelligence in nature inspire engineering designs?
3. Explain the importance of project-based learning in design engineering.
4. Discuss the role of life cycle design approach in design decisions.
5. Draw the figure of a smart phone which is both aesthetic and ergonomic
6. What are the aesthetic and ergonomic factors to be considered while designing the driver's seat for a passenger car?
7. Distinguish between project-based learning and problem-based learning in design engineering.
8. Describe how concepts like value engineering , concurrent engineering and reverse engineering influence engineering designs?

14 mark

1. Apply value engineering to a pen, and design a lightweight pen torch. Illustrate the solution using sketches.
2. Design waste bins to be kept at bus stops for waste collection enabling source separation. The bin should be theft-resistant and protect the contents of the bin from external weather conditions. Design the bins with ergonomic consideration for waste collection workers. Sketch the design using hand drawings.
3. Show the development of a nature-inspired design for a fashionable umbrella based on a banana leaf. Use hand sketches to support your arguments.
4. Develop some design modification for sports utility bag, to improve its functionalities as well as product value. Sketch the design
5. What is meant by modular design? Apply the modular design concept for a product- Bicycle.
6. Show the development of a nature inspired design for a solar powered bus waiting shed beside a highway. Relate between natural and man-made designs. Use hand sketches to support your arguments.
7. Show the design of a simple sofa and then depict how the design changes when considering 1) aesthetics and 2) ergonomics into consideration. Give hand sketches and explanations to justify the changes in designs.